Lecture 17

The Uncertainty Principle, Expectation Values

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Introduction

In lecture 16, we saw that a localized wave packet can be constructed by adding up (integrating) waves of various wave numbers k between

$$k - \frac{\Delta k}{2} \qquad \text{and} \qquad k + \frac{\Delta k}{2}.$$

The resulting wave was localized. We saw that if we choose bigger Δk, we get a more localized wave and vice versa.

$$\Delta k \propto \frac{1}{\Delta x}.$$

Today we look at this in a bit more detail.

In QM, sharply localized particles are replaced by waves. A particle with momentum p has an associated wave with

$$\lambda = \frac{h}{p}.$$

Therefore, any attempt to specify position and momentum must confront the fundamental limits set by wave behaviour.

To visualize this, consider an electron moving along x that passes through a slit of width D, i.e., $\Delta y \lesssim D$. That is we have reduced the uncertainty in its y-position coordinate to the slit width. We are sure that the electron passes throught the slit.

In the old Newtonian days, electron would emerge on the right side without change of its momentum: all momenta in x direction.

In Quantum Mechanics,

$$\lambda = \frac{h}{p}.$$

Such a wave diffracts and spreads out over an angle. That means it has some momenta in the y-direction as well.

$$\sin\theta \sim \frac{\lambda}{D}$$

$$\theta \sim \frac{\lambda}{D}$$

$$2\theta \sim \frac{2\lambda}{D}$$

$$\Delta p_y = p\sin(2\theta) \sim p\frac{\lambda}{D}$$

$$\Delta p_y \, \Delta y = p\lambda = h$$

Demonstrating that any attempt to localize the electron introduces uncertainty in its momentum.

Uncertainty Principle

Exact relationship between Δx and Δp depends on the shape of the wave and definitions of Δx and Δp.

Minimum value of $\Delta x\,\Delta p$ occurs for Gaussian waves with Δx, Δp defined as standard deviations of $\psi(x)$ and $\phi(k)$, its fourier transform.

$$\Delta x\,\Delta k = \frac{1}{2} \qquad \text{for Gaussian(bell-shaped) wavefunctions}$$

Generally,

$$\Delta x\,\Delta k \geq \frac{1}{2}$$

$$\boxed{\Delta x\,\Delta p \geq \frac{\hbar}{2}}$$

We started motivating this idea above based on lecture 16:

$$\Delta x \propto \frac{1}{\Delta k}$$

where we set $t = 0$ and calculated this.

Similarly, we would set $x = 0$ and show that

$$\Delta t \propto \frac{1}{\Delta \omega}$$

$$\Delta \omega\,\Delta t \gtrsim \frac{1}{2}$$

$$\boxed{\Delta E\,\Delta t \gtrsim \frac{\hbar}{2}}$$

Example

Electron in a hydrogen atom.

Atomic radius $= 5 \times 10^{-11}\,\mathrm{m}$.

$$\Rightarrow \Delta x = 5 \times 10^{-11}\,\mathrm{m}$$

$$\Delta p \geq \frac{\hbar}{2\Delta x}$$

$$\Delta p \gtrsim 1.1 \times 10^{-24}\,\mathrm{kg\,m/s}$$

Now that we understand a little bit about waves, let us start the process of finding the wave itself.

We already know

$$|\psi(x,t)|^2$$

is related to the probability of finding the particle at x at t.

Here $|\psi(x,t)|^2$ is the complex absolute value squared.

$$\int_{-\infty}^{\infty} |\psi(x,t)|^2\, dx = 1 \qquad \text{(Normalization)}$$

Besides being normalizable, the probability to find the particle between positions x_1 and x_2 is

$$P_{x_1,x_2} = \int_{x_1}^{x_2} |\psi|^2\, dx$$

Expectation Values

Once we find $\psi(x,t)$, we have probabilistic information. How do we get useful information from probabilities?

We calculate expectation values or averages.

Example: Six-faced die

Let us first recall some basic ideas about probabilities. Consider a 6 faced die, with the following probabilities.

$$P(1) = 0.20 = \frac{20}{100}$$

$$P(2) = 0.20$$

$$P(3) = 0.05$$

$$P(4) = 0.25$$

$$P(5) = 0.15$$

$$P(6) = 0.15$$

Average number coming up:

$$\langle x \rangle = \sum_x xP(x)$$

Toss the die $N = 100$ times (large number).

Average number coming up?

Let us say side i came up N_i times.

$$\text{Average} = \sum_i \frac{N_i}{N}\, i$$

$$= 1P(1) + 2P(2) + \cdots$$

$$\text{Average } i = \sum_i iP(i)$$

$$\boxed{\langle i \rangle = \sum_i iP(i)}$$

Continuous Variable

$$\langle x \rangle = \sum_x xP(x)$$

For a continuous variable,

$$\langle x \rangle = \int_{-\infty}^{\infty} dx\, xP(x)$$

$$\boxed{\langle x \rangle = \int_{-\infty}^{\infty} dx\, x|\psi(x,t)|^2}$$

or

$$\boxed{\langle x \rangle = \int_{-\infty}^{\infty} dx\, \psi^*(x,t)\, x\, \psi(x,t)}$$

This same procedure can be used to obtain expectation value of any quantity, like potential energy.

We can't however use this method to find expectation value of momentum p.

Because no function $p(x)$ can exist due to the uncertainty principle.

Hence we move to the machinery of operators.

Observable quantities in QM are represented by operators. That's for next time.